

Learner Guide

Retail Tracker — what to do, why, and where you are in the process | ★★★☆☆

This guide shows one sound way to complete the scenario, the reasoning behind each step, and which stage of the data analysis process each step belongs to. Work the task yourself first, then check your method and results here. Screenshots are illustrative mock-ups.

The data analysis process — follow it in order:

Identify → Collect → Clean/Prepare → Analyse → Visualise → Interpret/Report

Scenario 1 — Data Gathering and Validation

Step 1 — Identify and import

STAGE 1 · IDENTIFY & 2 · COLLECT

What you do: Import both CSVs, save as **TrendyThreads_Analysis.xlsx**, and confirm CustomerID is present and consistent in both files.

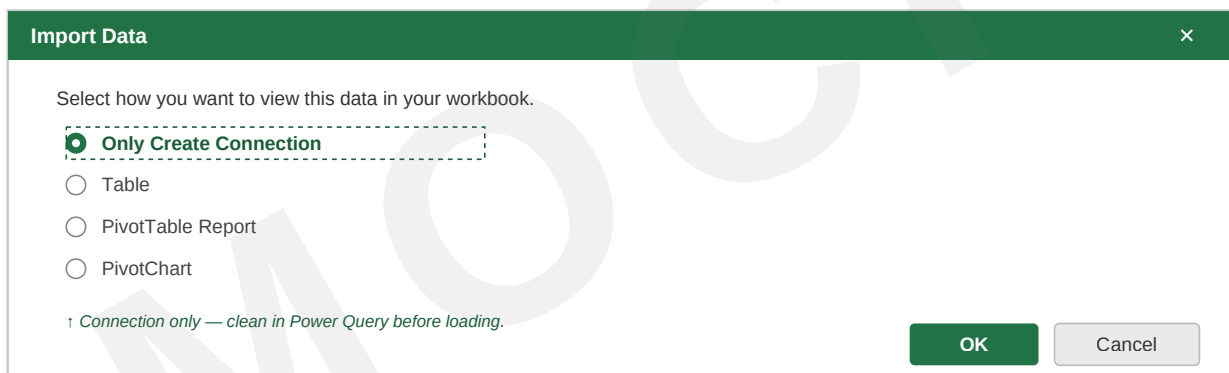


Figure 1 — Bring each file in as a connection-only query to clean before loading.

Why connection only? Even though these files are small and largely clean, using Get Data → Connection Only and double-clicking to open in Power Query keeps the process consistent with professional data-handling practice — and makes it easy to refresh if the source files update.

Step 2 — Remove duplicates and check dates

STAGE 3 · CLEAN / PREPARE

What you do: In CustomerProfiles, use **Data → Remove Duplicates** to remove the two repeated rows, then confirm dates in SalesData are all dd/mm/yyyy.

CustomerProfiles — duplicate CustomerIDs to remove

CustomerID	CustomerType	Region	Gender	Age	
C003	Online	South East	F	35	
C004	Store	Midlands	M	29	
C004	Store	Midlands	M	29	△ duplicate
C005	Store	Yorkshire	M	52	

C004 and C009 each appear twice. Data → Remove Duplicates resolves both.

After removal: 20 unique customer records matching the 20 sales transactions.

Figure 2 — C004 and C009 each appear twice; Remove Duplicates resolves both, leaving 20 unique records.

Why remove duplicates before blending? A duplicate CustomerID in the profiles table means XLOOKUP could pull the wrong record onto a transaction, or return inconsistent results. Cleaning first guarantees every lookup finds exactly one matching customer.

Step 3 — Blend with XLOOKUP

STAGE 3 · CLEAN / PREPARE

What you do: Add CustomerType and Region to SalesData using XLOOKUP on CustomerID:

```
=XLOOKUP([@CustomerID], Profiles[CustomerID], Profiles[Region])
```

H2	fx	=XLOOKUP([@CustomerID], Profiles[CustomerID], Profiles[Region])					
TransID	CustID	Category	Qty	UnitPrice	TotalCost	Region	CustType
T001	C001	T-Shirts	3	£15	£45.00	London	Online
T002	C004	Jeans	1	£45	£45.00	Midlands	Store

TotalCost: =[@Quantity]*[@UnitPrice]
XLOOKUP pulls Region and CustomerType from Profiles; TotalCost is Quantity × UnitPrice, formatted to currency.

Figure 3 — XLOOKUP brings Region and CustomerType across; TotalCost = Quantity × UnitPrice.

Why XLOOKUP? All 20 sales match a customer record, so no failed lookups here. XLOOKUP is preferred over VLOOKUP because it's not dependent on column position and handles exact matches cleanly.

Scenario 2 — Data Analysis

Step 4 — TotalCost column and Pivot Tables

STAGE 4 · ANALYSE

What you do: Add TotalCost, then build two Pivot Tables:

```
TotalCost: =[@Quantity]*[@UnitPrice] (format to currency, 2 dp)
```

Expected results:

Product Category	Total Sales
Shoes	£570.00 (top)
Jackets	£405.00
T-Shirts	£187.00
Jeans	£173.00
Accessories	£166.00

Average Order Value: Online **£74.30**, Store **£75.80** — almost identical, suggesting customer type has little bearing on how much is spent per order.

Step 5 — SUMIFS by region and HighValue? flag

STAGE 4 · ANALYSE

What you do: First use UNIQUE to list all regions, then SUMIFS to total TotalCost per region:

```
=SUMIFS(SalesData[TotalCost], SalesData[Region], [@Region])
```

```
HighValue?: =IF([@TotalCost]>100,"High Value","Standard")
```

Why UNIQUE first? Manually typing region names risks misspellings that will return zero — UNIQUE pulls the exact values that exist in the data, so every region is captured correctly.

Expected results: South West leads (£260), South East second (£240). **5 of 20** orders are High Value (>£100). Total revenue: **£1,501**.

Step 6 — Visualise

STAGE 5 · VISUALISE

What you do: Column chart of sales by category; pie chart of High Value vs Standard share; conditional formatting on the region table to highlight above-average regions.

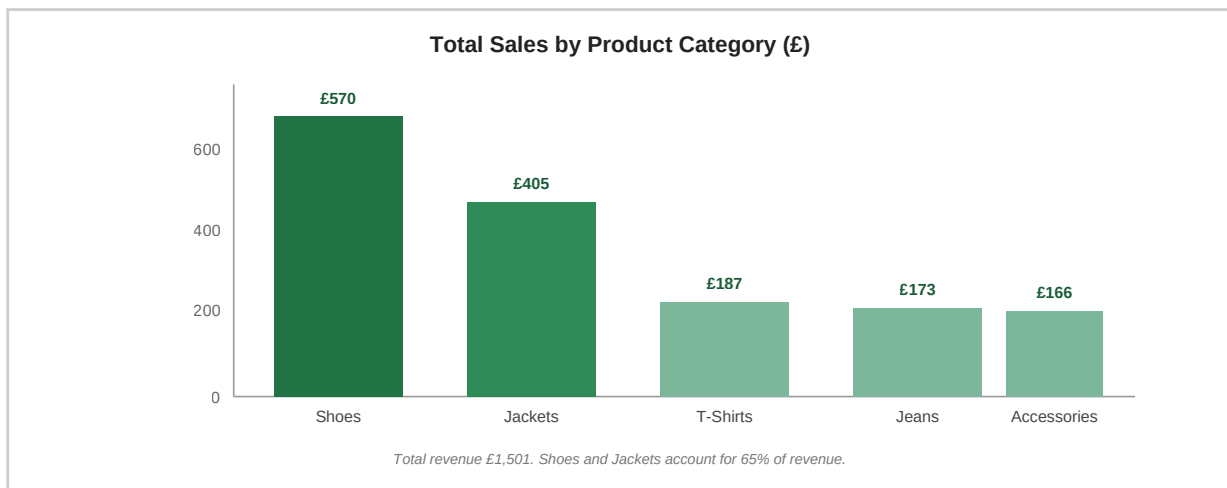


Figure 4 — Total sales by product category; Shoes and Jackets account for 65% of revenue.

Why conditional formatting on regions? The above-average regions aren't obvious from a table of numbers; a colour highlight makes the best-performing regions jump out immediately for a management audience.

Step 7 — Interpret and communicate

STAGE 6 · INTERPRET / REPORT

What you do: Write 3–4 sentences for senior management, identifying the top category, a note on customer type, and one recommended action.

"Shoes are the standout performer, generating £570 and accounting for 38% of total revenue; Jackets follow at £405. Online and Store customers spend almost identically (£74 vs £76 average), so the channel split does not explain revenue differences. South West and South East are the strongest regions. Trendy Threads should prioritise footwear stock and marketing in south-region stores ahead of the winter season, and investigate why Northern Ireland and North East are significantly lower."

Marker's checklist — process followed in order

- **Identify/Collect:** both CSVs imported, workbook saved, CustomerID confirmed as the key.
- **Clean/Prepare:** duplicate C004 and C009 removed; dates verified; blended with XLOOKUP (all 20 match).
- **Analyse:** TotalCost formula; two Pivot Tables; SUMIFS via UNIQUE; IF flag (>£100, 5 of 20 High Value).
- **Visualise:** column chart, pie chart, conditional formatting on region table.
- **Interpret/Report:** business-facing 3–4 sentence summary with a recommended action.